CE/CS/SE 3354 · Software Engineering · Fall 2026

Project Milestone P2

Analysis and High-Level Design

DigitIdentifier

Team	Team 05, section 010
Document version	1.0 (baseline for P2; grows from your P1 version 0.x)
Date	11/08/2026
Repository URL	https://github.com/drishyakoirala/Team5-SoftwareEngineering
Git tag for this milestone	p2 (points at the commit that contains this exact PDF)
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0.1 Revision History, Contribution Record and Changes Since P1

Version	Date	Author	Summary of change
0.1	10/15	Drishya Koirala, Prakriti Mehta, Sho Shimizu, Telile Tura, Mehrsa Hosseini	Initial baseline for P1
0.2	11/08	Drishya Koirala, Prakriti Mehta, Sho Shimizu, Telile Tura, Mehrsa Hosseini, Xiangrui Teng	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Drishya Koirala	I completed section 4.1, 4.2, and 4.3 as well as section 8 open question Q01 in section 11.	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit
Sho Shimizu	I completed section 6.4, 6.5, 6.6 and Q05 on section 11	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit
Telile Tura	I completed section 6.1, 6.2, 6.3, and open question Q02.	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit
Prakriti Mehta	I completed section 7.1 and 7.2 and open question Q03	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit
Xiangrui Teng	I completed section 5.3 and 9.1-9.2	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit
Mehrsa Hosseini	I completed section 5.1, 5.2, and open question Q06 in section 11	Commit number: 0922355, 652295, a91c47a, 859182c, f8ac8c5, edd6466, 4986af7	Has reviewed commit

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	Spoke to a teacher in order to solidify the use cases and the functionality of the program. Found supporting evidence for stakeholder needs and feasibility (C1)	P1 feedback identified a mismatch between stakeholders and project scope and requested checkable evidence for feasibility assessments.	Makes the stakeholder analysis and feasibility assessment more consistent and defensible.	11/08 - P2
CH-02	Added measurable thresholds to QR-03 and QR-04 and introduced a privacy requirement (C2).	P1 feedback identified as vague and untestable. Stated that privacy considerations were missing for sensitive psychologist responses.	Makes requirements testable and establishes privacy expectations.	11/08 - P2
CH-03	Updated the GitHub repository and working agreement with project documentation, contribution records, folder structure, and communication expectations (C4).	P1 feedback identified insufficient repository content and missing team communication procedures.	Improves project organization, contribution visibility, and team coordination.	11/08 - P2

1 Introduction

1.1 Purpose of this document and of the system

This document serves as the starting point for the DigitIdentifier project and is intended for the team creating the product, stakeholders, and other individuals involved in the development and evaluation of the system. DigitIdentifier is a system designed to recognize handwritten digits from 0–9 and convert them into usable digital data. The project will explore two major use cases such as number identification in a scantron as well as a method for psychologists to gather data in the pen and paper style yet have digitized data which they can further manipulate themselves.

DigitIdentifier allows for the identification of numbers 0-9 written down on paper. Due to the vast nature of possible use cases, we plan to create a model which can later be further developed to be implemented into scantron machines to eliminate the bubbling process of numbers in the number grid. In terms of scantron use, the program that we develop will not have the tools to integrate it into the scantron or the abilities for it to be compatible with a scantron. The scope of this project is to create a program which can identify a number written on a piece of paper. We will also plan to have the model be used by psychologists to get workable metrics from data collected whilst conducting studies. Many students complain about the concept of having to bubble in the number in the number grid of the scantron after writing the number down, the concept of having to bubble in each number can be eliminated with the integration of DigitIdentifier. Psychologists to this day continue to use pen and paper methods to gather statistics for observational studies due to its reliability, but would definitely benefit from a digitized version of their data. Both of these problems will be addressed with the DigitIdentifier.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Recognition of handwritten digits 0–9 from images of numbers written on paper	Recognition of anything other than digits, this includes letters, decimal points, any other symbols, etc. The project's scope is to only recognize digits.
A machine-learning model that identifies the digit shown in an input image.	Recognition of a number with multiple digits. The project's scope is to only recognize standalone digits, thus the recognition of a number with multiple digits will not be implemented.
A system that provides the predicted digit and its confidence score	Integration with scantron machines and providing summarized data or any order of data manipulation after the digit is recognized. The project is only providing a digit recognition capability not an integration capability for scantron machines.
Preprocessing input images and producing a predicted digit with a confidence score which gives insight on accuracy.	This project will not support any other sort of input other than input from a usb connected camera or the laptop camera. The project is not designed to have any other sort of input.

The excluded contents are excluded because they do not follow the scope of the overall project. We have decided to exclude the recognition of anything other than digits because that simply would no longer provide for the possible use cases we have intentions for. We also decided to exclude the recognition of numbers with multiple digits because this program is to aid the recognition of numbers on a 0-9 scale, not to read numbers with multiple digits. We are not creating an integratable software with scantron machines as that is simply out of the scope that we have created. Finally, we will not accept any other form of input other than usb connected camera and laptop camera.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Image Preprocessing	Manipulating the image to fit the requirements of the program which will recognize the digit
Confidence Score	Probability on how confident the model is at predicting the digit.
Predicted Digit	This is the digit that DigitIdentifier identifies and returns to the user.
Input Image	Image of hand written number that the user provides for the DigitIdentifier.

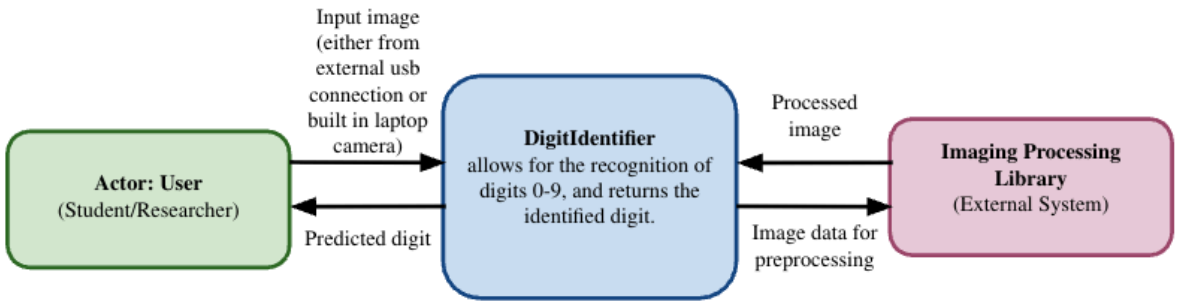
1.4 References

[R1]

Virues-Ortega, J., Casas, C. D., Martin, N., Tarifa-Rodriguez, A., Hidalgo, A. J. R., Cox, A. D., & Navarro Guzmán, J. I. (2023, February). *Accuracy of paper-and-pencil systematic observation versus computer-aided systems*. Behavior research methods. https://pmc.ncbi.nlm.nih.gov/articles/PMC10027644/

2 Overall Description

2.1 Product perspective and context



The boundary sits where it does because the scope of this project is to create a program which can take 2 possible inputs which are either external usb connected cameras which will be shown at the paper with the written number, or the computer's built-in camera which will be shown the paper with the written number. Thus, making the input from the actor which is the user to the DigitIdentifier to be simply an input image which has two possibilities on how it is passed over. The Imaging Processing Library which will be decided further down the line will be an external system which we will integrate into our program in order to do the preprocessing for the image. The library will then hand the processed image to DigitIdentifier for it to then identify the digit which will then return the predicted digit to the actor which is the user. This covers the overall scope of this project and what will be used as input and what will be given as output to the actor.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Teachers	Primary users	fast, easy, and accurate grading	Teachers don't have a lot of time and with multiple students in each class, they have many exams that need to be graded. It is important that each exam is graded accurately to ensure fair grading policy.
Students	directly affected by results	accurate and fair exam scores	Grades are a huge determining factor to students' performance in their classes. They can determine their future and many students put a lot of work into keeping their grades up. Since exam scores have a lot of weight in the gradebook, it is vital that they accurately portray a student's performance.
School administrators	help with school decision making	reliable application that is accepted by teachers and students	Administrators are responsible for student safety and protecting student information. They also care that student grades are calculated based on academic performance to uphold academic integrity. They also care that staff, like teachers, have a comfortable experience when teaching, grading, etc.

Stakeholder	Role in relation to the system	What they need from it	How you learned this
School district	purchase software	good value and reasonable cost, ethical software	School districts have a set budget for each school year. They need to distribute their funding to pay for teachers, school supplies, computers, software, building maintenance, and many other expenses. When school administrators want to adopt new software, they require the approval of the district who will approve the purchase and adoption. The district will consider the costs and policy of the application to make sure it is in line with district policy.
Developers/Software company	build the software	understand all school/policy requirements and needs	Software Developers work to fulfill the needs of their customers. They need to clearly understand the software requirements (like what features to include or leave out) in order to meet customer satisfaction.
Maintainers	update and repair software after release	well documented code; tools and procedures to fix bugs and integrate updates with ease	To make sure the app will continue to work as intended and improvements are made to keep up with demands, maintainers need to access the app's code and work on already existing software. well documented code will keep everything organized and help avoid unnecessary work such as deciphering complicated lines of code.

2.3 User characteristics

Teachers: For in-class assignments or tests, teachers commonly use a simplified answer sheet and use a scanner to quickly grade assignments. It is likely that they are familiar with digital grading tools but not as knowledgeable with image recognition software. Our software should allow them to scan forms and review the results.

Students: Students will be filling out the scantrons with their own handwriting, which will drastically vary per student. They are not expected to know how the scanner software recognizes their answers. The answer sheet needs to provide clear instructions and sufficient space for students to write legibly.

Test Administrators: Test administrators must distribute, collect, and process answer sheets. They may have to process a large amount of answer sheets and ensure that it corresponds with the correct participant. The scanning software must provide straightforward instructions and inform the user about any errors during the process.

Psychologists: Psychologists may use the scanner software during screening processes to digitize responses. They may not be familiar with image recognition software. The responses they obtain also contain sensitive and personal information. The software must ensure confidentiality of responses and flag any uncertainty during the scanning process.

2.4 Constraints

The software only recognizes digits 0-9. Letters or symbols will not be recognized. Notably, decimal format is not supported.

The software can only recognize handwritten digits within a specific space or box.

The scantrons must be versatile enough so that it can be used for a variety of fields, from teachers to psychologists as described in 2.3.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Users have access to a camera to readily take pictures of their scantron and run our software.	Users will be unable to capture images of the scantrons, which will prevent them from using our software unless an alternative camera or image-upload method is provided.
A2	Users will write single digit responses clearly in designated boxes.	The software may misread digits and produce incorrect grades or quiz scores.
A3	The scantrons will follow a consistent layout with the handwritten digits placed in clear locations.	The software may read digits from incorrect locations or associate responses with the wrong questions.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	conditionally feasible	A team member has completed a similar project using a scanner to read single digit numbers. Although this is a different project, the same framework is likely to be helpful.
Operational	feasible	With a working model, schools and teachers are expected to give the platform a try.

Feasibility gate	Verdict	Evidence behind the verdict
Schedule	conditionally feasible	With this being a new project idea, we are unsure of the hurdles and setbacks that are expected with the software application.
Economic / resource	feasible	We expect low level costs: Aside from team member time, there are no additional costs at the moment.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall scan input containing single digit numerical responses	MUST	end user	A tester can submit a supported survey form and verify that the system scans it successfully
FR-02	The system shall identify and recognize single digit numbers marked or written in designated response areas	MUST	end user	Given a test form with known numerical responses, the system detects the number in the designated response areas
FR-03	The system shall record each detected number as a digital value	MUST	end user	After scanning a test form, the tester can verify that each recognized number is recorded as the corresponding digital value
FR-04	The system shall identify response areas where no number is detected	SHOULD	end user	Given a tests form containing blank response areas, the system flags those areas as having no detected response
FR-05	The system shall flag responses that cannot be recognized for manual review	COULD	end user	Given an unclear/unreadable response, the system flags the response instead of just assigning a number
FR-06	The system shall allow user to correct incorrectly detected numbers	COULD	end user	A user can change an incorrectly detected value, and the corrected value is changed in the saved results
FR-07	The system shall save detected responses	WONT	end user	After scanning the tester will be given the recognized digit user can save the results at their own discretion.
FR-08	The system shall maintain order of responses	WONT	end user	Given an input the model will not record all the values in that exact order. That is up to the user and they can save values at their own discretion.

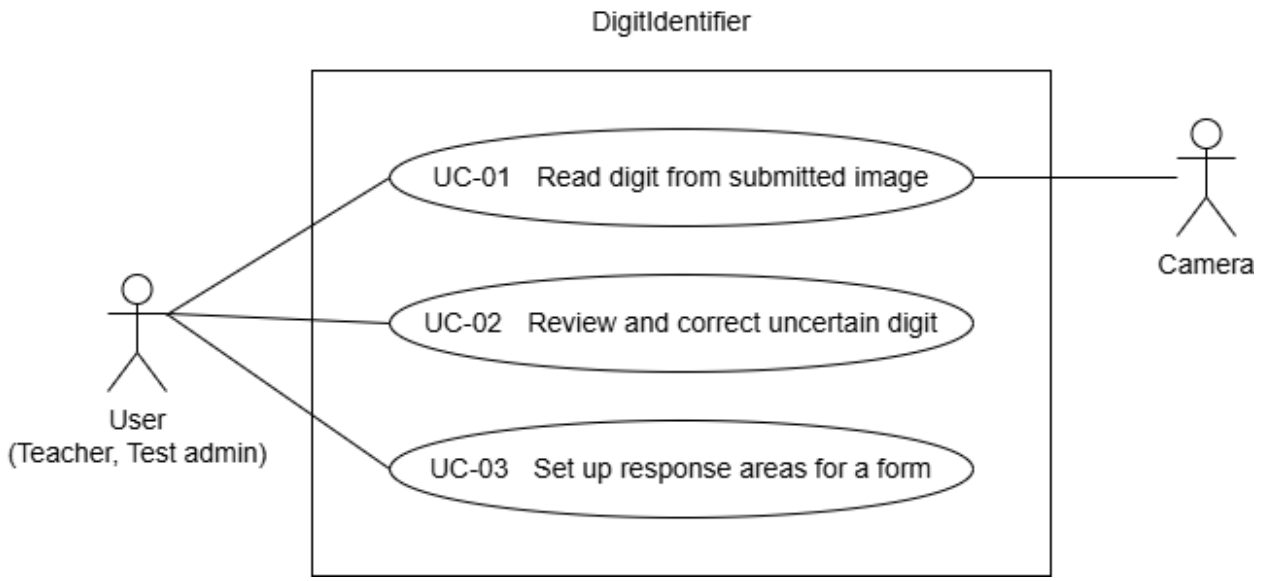
3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	The system shall correctly recognize at least 95% of clearly written single digit responses under normal scanning conditions, measured manually from 100 test responses	MUST	end user	Scan 100 responses and compare detected values with verified answers
QR-02	The system shall process an input containing 50 response fields within 10 seconds under normal operating conditions, measured from submission scan to displayed results	COULD	end user	Scan input with 50 response fields and measure the processing time
QR-03	The system shall display scanning results within seconds after processing is complete, measure by timing over 10 test scans	MUST	end user	Measure the time between processing completion and results appearing on screen
QR-04	The system shall display scanning results in a readable format for every successfully scanned survey	MUST	end user	Scan 5 test surveys and verify that the detected numbers are clearly displayed and readable

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall only detect and record single digit numerical responses and shall not interpret psychological test results.	Project scope/stake holder agreement	The agreed project scope is limited to number detection not psychological analysis
CON-02	The system shall support survey responses within the configured scale such as 1-5 or 1-10.	end user/project requirements	The system must accommodate the numerical response formats required by its intended users

3.4 Initial use cases



Expanded use case

Field	Content
Identifier and name	UC-01 Read digit from submitted image
Primary actor	User (teacher, test administrator)
Supporting actors	Camera
Preconditions — what must be true before it starts	UC-03, Response area for the form has to be set up Camera is connected to device and app is running
Trigger — what starts it	User have a form they want to make it to digital value
Main success scenario — numbered steps	1. User places form in front of camera 2. FR-01, System captures image from camera 3. FR-02, System locates designated response area 4. FR-02/FR-03, System identify the digit in response area with its confidence score 5. FR-04/FR-05, System marks area with no digit as blank, and flag digit with not enough confidence score 6. QR-03/QR-04, System display every response area with its digit and confidence score, highlight blank and flags 7. The user gets the displayed value.
Alternative and exception flows	2a. The image is too dark, blurry, etc. and can't be read: System asks user to retake and return to step 1 3a. system can’t locate the response area: system asks user to retake and return to step 1

Field	Content
	4a. CON-01, Response area have 2 digits/letter: response returns low confidence and flagged at step 5 6a. one or more readings are flagged: user continues with UC-02
Postconditions — what is true when it succeeds	Every response area on the form is displayed as a digit with its confidence score, blank, or flagged for review.
Requirements realised	FR- 01, FR-02, FR-03, FR-04, FR-05, QR-01, QR-02, QR-03, QR-04

Expanded use case

Field	Content
Identifier and name	UC-02 Review and correct uncertain digit
Primary actor	User (teacher, test administrator)
Supporting actors	None
Preconditions — what must be true before it starts	UC-01 has returned results, and at least one reading is flagged or looks wrong to user
Trigger — what starts it	User sees flagged/incorrect result and wants the value to be corrected
Main success scenario — numbered steps	1. User selects flagged reading 2. System shows the image of that response area with its predicted digit and confidence level 3. FR-06, User enters the correct digit 4. System replaces predicted digit with entered digit and clears flag 5. Repeat step 1-4 until no flag is remaining
Alternative and exception flows	1a. FR-06, Reading is wrong but not flagged: User manually selects it and corrects it the same way 3a. CON-01User enters other than single digit: System rejects entry and keeps flag 3b. CON-02, The digit is outside the form's configured scale: system warns user and asks for confirmation 3c. The response is unreadable to user: User marks it blank
Postconditions — what is true when it succeeds	Every displayed value is confirmed by the user.
Requirements realised	FR- 05, FR-06, CON-01, CON-02

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	UC-01	
FR-02	UC-01, UC-03	
FR-03	UC-01	

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-04	UC-01	
FR-05	UC-01, UC-02	
FR-06	UC-02	
FR-07		Not for this release since marked as WONT
FR-08		Not for this release since marked as WONT

Requirements were mostly completed by UC-01 and few on UC-02/03, but there was no missing use case or requirement. FR-07/08 won't be implemented in this release.

4 Use-Case Analysis

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Read digit from submitted page	This is the main purpose of the application: identifying handwritten digits from an input image	Capture an image, preprocess the image, identify the digit, and display the final prediction.
UC-02	Review and correct uncertain digit	Recognition may not always be accurate due to the user submitting a blurry image, so users should be able to review predictions when they feel as though it's inaccurate due to user error.	Check the confidence score, if the user made an error, allow the user to correct the digit, and update the result.
UC-03	Set up response areas for a frame	The program needs to identify and separate individual digits within an image before recognizing them	Detects digit locations, separate individual digits, create image regions for each digit, and prepare them for recognition.

4.2 Expanded use cases

UC-01 Read digit from submitted page

Primary actor	Teacher/Psychologist
Supporting actors and external systems	Camera and NumberIdentifier program
TUCBW (this use case begins when)	This use case begins when the Teacher/Psychologist begins scanning a handwritten digit.
TUCEW (this use case ends when)	This use case ends when the program displays the recognized digit.
Preconditions	The program is running, the camera is available, and a handwritten digit is visible.
Postconditions	The digit is recognized and displayed to the Teacher/Psychologist.
Traces to requirements	FR-01, FR-02, FR-03, QR-01, QR-03, QR-04, CON-01, CON-02

Actor input / action	System response
	(0) Program presents user with a platform to capture image
(1) The user positions the paper with the handwritten digits in cameras view	(2) The program processes the image
(3) The user submits the captured image for recognition	(4) The program preprocesses the image and identifies the handwritten digit
(5) User waits for program response	(6) The program displays the predicted digit and confidence score

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	2	No image is captured	Program is unable to give a predicted digit	Step 1
A2	4	Digit is not able to be recognized	Program will return a digit with a low confidence score, which in turn the user has a chance to recapture an image	Step 3

UC-02 Review and correct uncertain digit

Primary actor	Teacher/Psychologist
Supporting actors and external systems	NumberIdentifier program
TUCBW (this use case begins when)	The program displays a digit with a low confidence score.
TUCEW (this use case ends when)	The Teacher/Psychologist confirms or corrects the predicted digit.
Preconditions	The program has scanned a handwritten digit and generated a prediction.
Postconditions	The uncertain response is reviewed and corrected, and the predicted digit is confirmed or corrected by the Teacher/Psychologist.
Traces to requirements	FR-05, FR-06, QR-04, CON-01, CON-02

Actor input / action	System response
	(0) The program identifies a digit with a low confidence score

Actor input / action	System response
(1) User reviews predicted digit	(2) The program displays predicted digit with confidence level
(3) User reattempts by submitting another captured image for recognition	(4) The program updates the recognized digit
(5) The user confirms the result	(6) The program displays the new confirmed digit

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	3	User is satisfied with the predicted digit	The user confirms the original predicted digit without changes.	Step 2
A2	3	User is unable to determine the correct digit	user has a chance to recapture an image	Step 6

UC-03 Set up response areas for a frame

Primary actor	Teacher/Psychologist
Supporting actors and external systems	Camera, NumberIdentifier program, image processing component
TUCBW (this use case begins when)	The camera captures a frame containing handwritten digits
TUCEW (this use case ends when)	The program identifies and prepares individual digit areas for recognition
Preconditions	The program is running, the camera is available, and handwritten digits are visible.
Postconditions	Individual digit regions are identified and prepared for recognition.
Traces to requirements	FR-01, FR-02, FR-04, QR-01, QR-02, CON-01, CON-02

Actor input / action	System response
	(0) The program receives a camera frame
(1) The user positions the handwritten number such that it is in cameras view	(2) The program processes the frame and identifies potential digit locations
(3) The user waits for processing to finish	(4) the program separates regions that contain digits from regions that do not
(5) N/A	(6) N/A

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	2	Digit is not detected in given frame	Program should prompt user that there is not anything in the frame	Step 1
A2	4	Multiple digits in one region, and program is confused which one is supposed to be read	Program should prompt user that image is not suitable	Step 1

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-01	☑ ~~Is the actor a ROLE, not a person, a department or "user"?~~ ☐ Is there a step (0) saying what the actor finds before acting? ☑ ~~Does every row hold exactly ONE turn?~~ ☑ ~~Does every item the actor supplies exist as a domain model attribute?~~ ☑ ~~Does the use case end with an acknowledgment the actor can perceive?~~ ☑ ~~Are the pre-conditions things you may ASSUME, not failures you must handle?~~ ☑ ~~Is there a post-condition a tester could turn into a test?~~ ☑ ~~Is every alternate named by the step it branches from?~~ ☐ Is there any background processing that should not be here? ☐ Does any row name a technology, a store or a screen widget?	The initial version did not clearly describe what the actor sees before scanning. It also included internal image-processing steps.	Added step (0) to show that the program is ready to scan and remove unnecessary internal processing details.

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-02	☑ ~~Is the actor a ROLE, not a person, a department or "user"?~~ ☐ Is there a step (0) saying what the actor finds before acting? ☑ ~~Does every row hold exactly ONE turn?~~ ☑ ~~Does every item the actor supplies exist as a domain model attribute?~~ ☑ ~~Does the use case end with an acknowledgment the actor can perceive?~~ ☑ ~~Are the pre-conditions things you may ASSUME, not failures you must handle?~~ ☑ ~~Is there a post-condition a tester could turn into a test?~~ ☑ ~~Is every alternate named by the step it branches from?~~ ☐ Is there any background processing that should not be here? ☐ Does any row name a technology, a store or a screen widget?	The initial version did not specify what the actor sees before reviewing an uncertain response. It also included internal confidence-checking details	Added step (0) to show the uncertain response and focused the steps on the actor reviewing and correcting the digit.
UC-03	☑ ~~Is the actor a ROLE, not a person, a department or "user"?~~ ☐ Is there a step (0) saying what the actor finds before acting? ☑ ~~Does every row hold exactly ONE turn?~~ ☑ ~~Does every item the actor supplies exist as a domain model attribute?~~ ☑ ~~Does the use case end with an acknowledgment the actor can perceive?~~ ☑ ~~Are the pre-conditions things you may ASSUME, not failures you must handle?~~ ☑ ~~Is there a post-condition a tester could turn into a test?~~ ☑ ~~Is every alternate named by the step it branches from?~~ ☐ Is there any background processing that should not be here? ☐ Does any row name a technology, a store or a screen widget?	The initial version included internal image-processing steps and referenced implementation components rather than focusing on the actor's interaction with the system.	Added step (0) to describe the available survey form and simplified the steps to focus on identifying and preparing response areas.

5 Analysis Models

5.1 Domain model

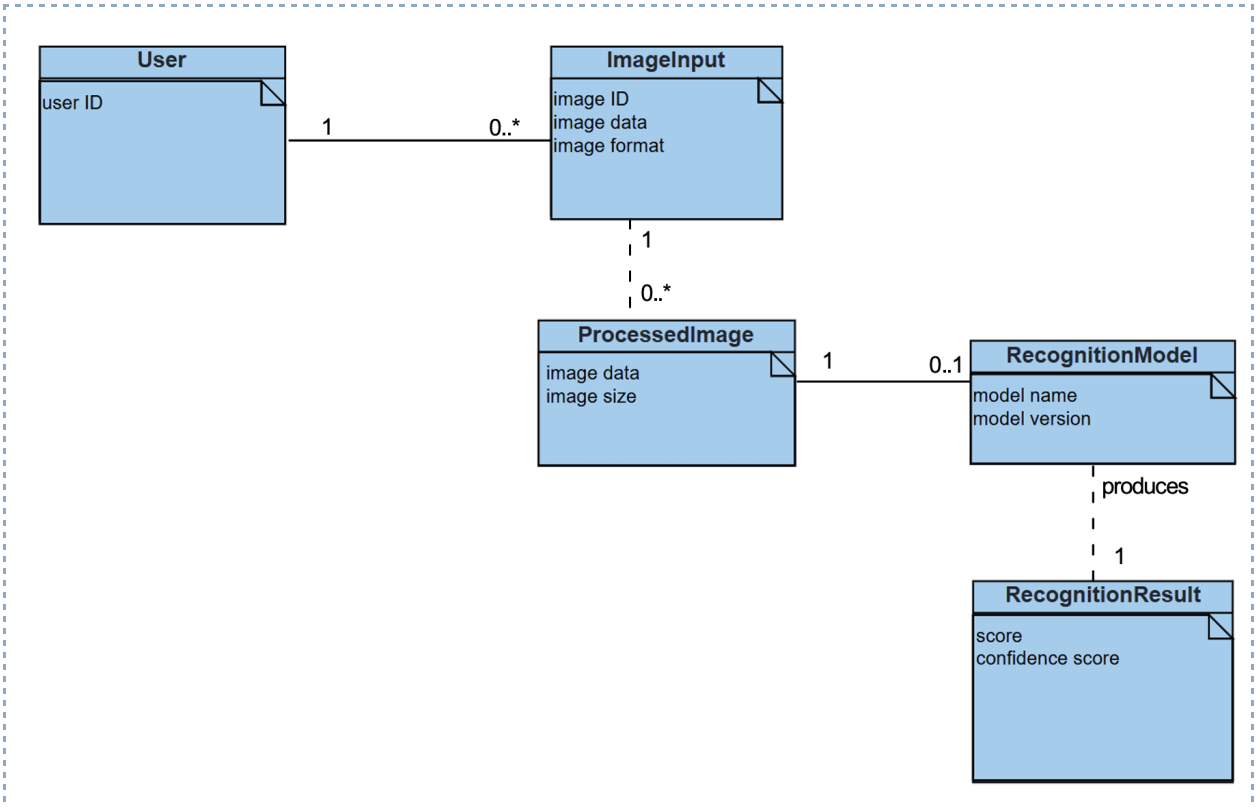


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
User	The person who interacts with the app and takes the picture of the scantron.	User ID	A user takes 0..* ImageInputs Each ImageInput belongs to 1 User
ImageInput	The original image that was captured by the user.	Image ID, Image Data, Image format	A ImageInput produces 0..1 ProcessedImage Each ProcessedImage comes from 1 ImageInput
ProcessedImage	The cleaned image that has been prepared from the original input.	Image data, Image size	A ProcessedImage produces 0..1 RecognitionResult Each RecognitionResult comes from 1 PrecosedImage

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
RecognitionModel	The training model responsible for identifying the correctness of the scantron questions from number	Model version, Model name	A recognitionModel produces 0..* RecognitionResults Each result comes from 1 RecognitionModel
RecognitionResult	The outcome of the recognition model with prediction for the score results	Result ID, Score, Confidence score percentage	Each RecognitionResult comes from 1 PrecessedImage And 1 RecognitionModel

5.2 Object interaction models

5.2.1 UC-01 Read digit from submitted page

(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.

(0) The program is ready to capture an image : **trivial**, program is simply waiting for user input

(1) The user positions the paper with the handwritten digits in cameras view : **trivial**, the user physically positions the paper with no system interaction

(2) The program processes the image : **non-trivial**, the system must perform analysis to process the image

(3) The user submits the captured image for recognition : **trivial**, the user simply initiates the recognition process

(4) The program preprocesses the image and identifies the handwritten digit : **non-trivial**, the system must prepare the image and perform digit recognition

(5) User waits for program response : **trivial**, waiting does not involve an action or calculation

(6) The program displays the predicted digit and confidence score : **non-trivial**, the program must perform analysis to find confidence score and fetch/display results

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	User	capture an image	Camera/phone	Image Processing software
2	ImageInput	preprocess the image	Standard Image data	Image Processing software
3	Digit Recognition Model	identify the digit	Processed Image data	RecognitionResult
4	Recognition Results	display the final prediction	Recognized digits	User

5.2.2 UC-03 Set up response areas for a frame

(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.

(0) The program receives a camera frame : **non-trivial**, system must capture and process the input

(1) The user positions the handwritten number such that it is in cameras view : **trivial**, the user physically positions the camera (no internal analysis)

(2) The program processes the frame and identifies potential digit locations : **non-trivial**, the system must analyze the image

(3) The user waits for processing to finish : **trivial**, the user waiting does not involve interaction with system

(4) the program separates regions that contain digits from regions that do not : **non-trivial**, the system must dissect the image to do so

(5) N/A : trivial, no action is required

(6) N/A : trivial, no action is required

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Image Processing software	detects digit locations	Standard Image data	Image Processing software
2	Image Processing software	separate individual digits	Processed Image data	Image Processing software
3	Image Processing software	create image regions for each digit	Processed Image data	Image Processing software
4	Image Processing software	prepare image regions for recognition	Processed Image data, Recognized digit areas	User

(c) Analysis sequence diagram.

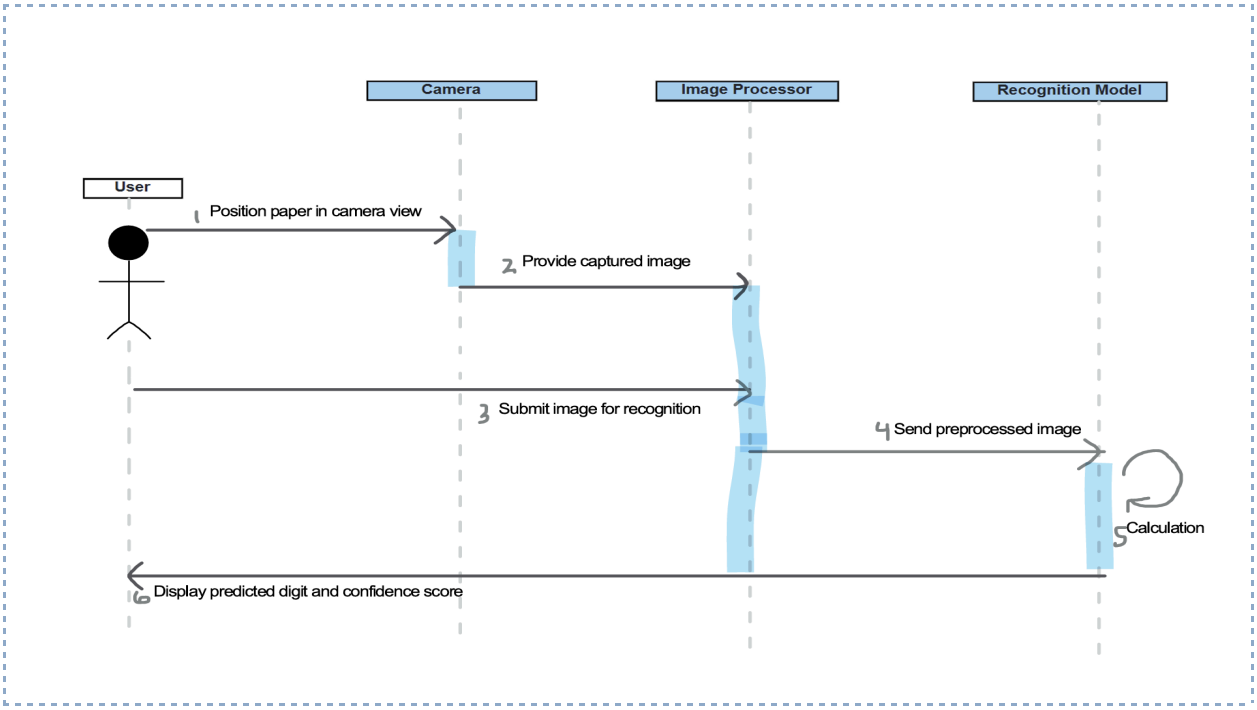


Figure 5.2.1 Analysis sequence diagram for UC-01; message numbers match the scenario table

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Partial. The selected response area and corrected digit lack explicit attributes.	Open(5.1)
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Partial. Camera and Image Processor need explicit interaction-object definitions.	Open(5.2)
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Fail. UC-01 has six diagram messages but for scenario rows.	Open(5.2.1)
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	Partial, UC-03 step 0 has no frame-capture now	Open(5.2.2)
K5 Every analyzed use case traces to at least one functional requirement	Pass. UC-01, UC-02 and UC-03 trace to FR-01, FR-06 and FR-02 respectively.	Verified(4.2)
K6 The last messages of each diagram make the use case postcondition true	Partial.UC-01 displays the digit and confidence.UC-03 still needs a diagram	Open(5.2.2)

6 High-Level Design (Architecture)

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Accuracy	The user submits a scantron containing 100 handwritten single digit responses.	Digit recognition program	The system processes the image and identifies each handwritten digit	At least 95 of the responses must be read accurately by the system.	QR-01
QAS-02	Performance	The user finishes scanning a survey with on a scantron	Result display interface	The system displays the processed results to the user	A test run of 10 scans should each display results within 5 seconds after processing scans.	QR-03
QAS-03	Readability	The user views results after scanning.	Result display interface	The system legibly displays the recognized numbers in a clear, organized format	At least 95% of the displayed digit responses are fully visible and accurately associated with their respective questions across 5 different scans.	QR-04

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Layered	QR-01, QR-03, QR-04. Separates image processing, digit recognition, and results display into their own layers	Defining communication between layers and changes to one layer may require changes to another.	Chosen
Pipe-and-filter	QR-01. Allows images to go through separate processing stages, such as preprocessing and digit segmentation	Can cause complications when sharing information between stages.	Rejected

Style considered	Drivers it serves	What it costs	Verdict
Client-server	Usability. Allows users to submit images through a client while the server handles the recognition and processing.	Requires network communication and server infrastructure.	Rejected

Our team decided on a layered architecture style because it creates distinct components for the system, such as image capture, image processing, and results display. It best supports our accuracy requirement by allowing each stage to be developed and tested independently. It supports our performance and readability requirements by separating the processing and recognition from the results display. The pipe-and-filter was rejected because it was less suited to managing user interaction and results. The client-server was rejected because our prototype does not require separate server or network communication.

6.3 Component view

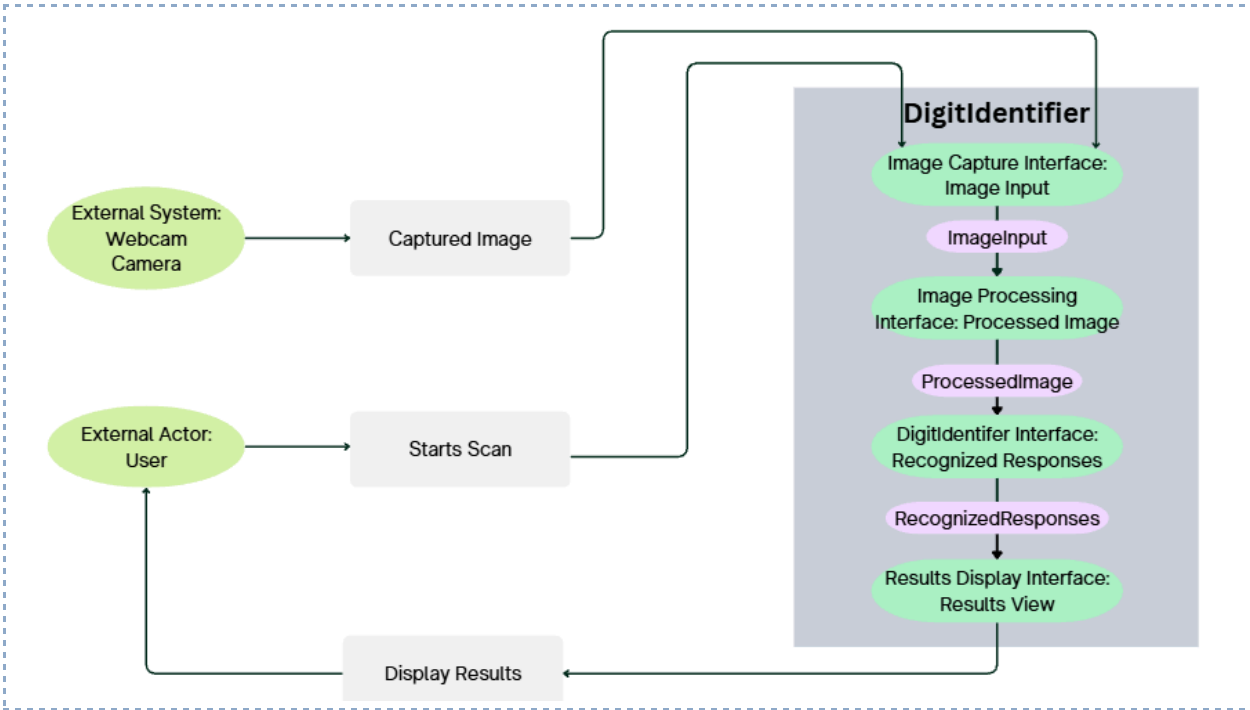


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Image Capture	Captures an image of a completed scantron.	ImageInput	Webcam camera	UC-01, ADR-01

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Image Processing	Processes the captured image by locating response areas and adjusting the image for recognition	ProcessedImage	Image Input	UC-01, QAS-01, ADR-01
Digit Recognition	Identifies handwritten digits in response areas.	RecognizedResponses	ProcessedImage	UC-01, QAS-01, ADR-01
Results Display	Shows the recognized digits in readable format for user	RecognitionResult	Recognized Responses	UC-02, QAS-02, QAS-03, ADR-01

6.4 Deployment view

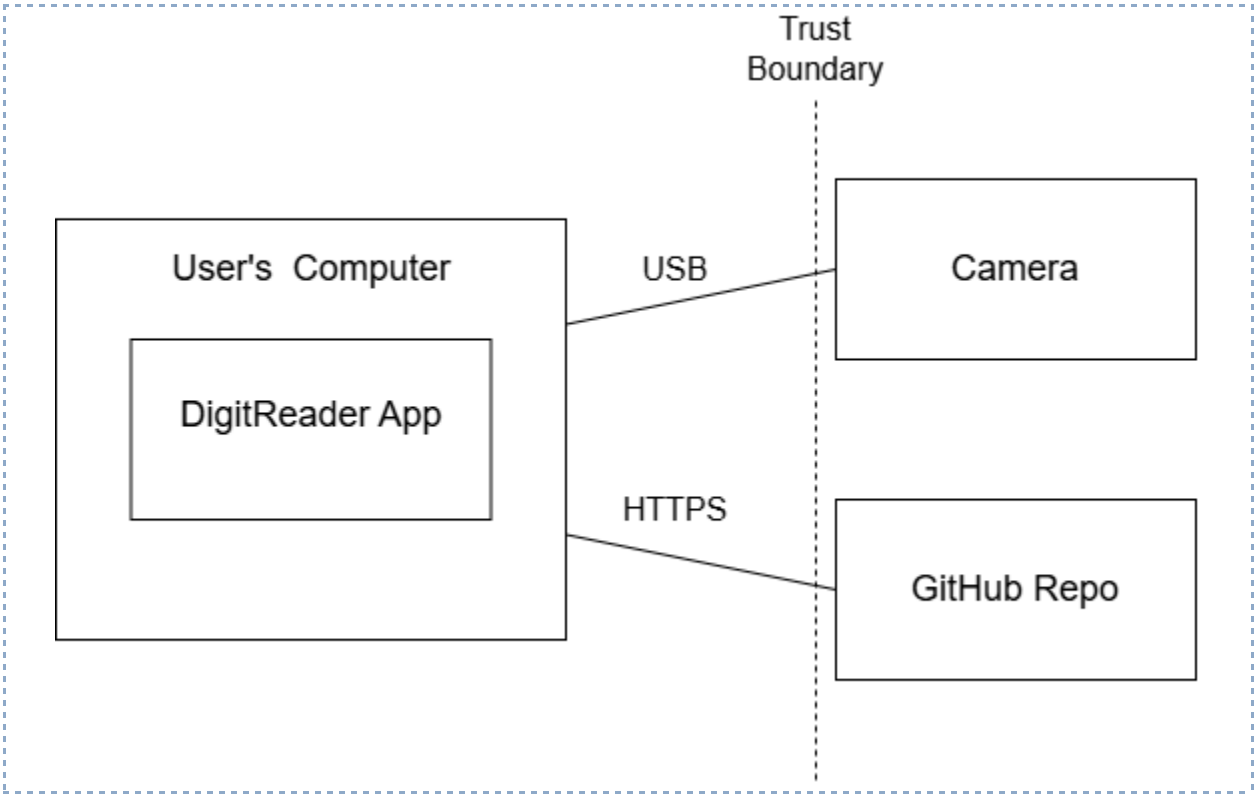


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
User's Computer	Runs application of DigitReader	Camera by USB GitHub by HTTPS	Trusted

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Camera	Only the camera's firmware	User's Computer by USB	Outside trust zone
GitHub Repo	Hosts the actual code of DigitReader	User's Computer by HTTPS	Outside trust zone

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
User	Results Display	External, uses the system through screen
Camera	Image Capture	External, outside the trust boundary
ImageInput	Image Capture	Created by Image Capture
ProcessedImage	Image Processing	Created by ImageInput
Image Processing Software	Image Processing	Created by ImageInput
Digit locations	Image Processing	Merged, prepare digit before recognition
RecognitionModel	Digit Recognition	Recognize the digit after Image Processing
RecognitionResult	Digit Recognition	Created by Digit Recognition
Read Cycle	Results Display	Merged, display the recognized digit
Review of uncertain digit	Results Display	Flag and resubmit the low confidence score image

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Capture, preprocessing, recognition and display are separate layers and each handles one aspect of reading digit.	6.2 layers
Information hiding	Each component hides one feature from the other component. Image capture hides the camera, image processing hides the algorithm, digit recognition hides the model file, and results display get the response from digit recognition instead of raw output.	Image capture, Image processing, digit recognition
Low coupling	Each layer passes only the values to the next layer, such as digit and percentage and not the raw data.	Image capture, Image processing, digit recognition
High cohesion	image processing turns input to digit images, digit recognition decides which digit image shows, with percentage.	6.5 Image processing, digit recognition
Design for change	Likely changes are camera upgrade and better lighting handling. Which changes single component inside Image Processing	Digit Recognition, Image Capture, Image Processing
Keep it simple (KISS)	One sentence explains the deployment. One program on the computer reads the camera and shows the digit. There's no server, database, or network. The results are shown but not stored.	6.2 rejected client-server

7 Design Rationale

7.1 Architecture decision records

ADR-01 Use a layered architecture

Status	Accepted
Date and owner	November 8, 2026
Context (drivers)	QR-01, QR-03, QR-04. The system needs to process uploaded images, recognize digits accurately, and display results in an organized way. Separating these responsibilities makes the system easier to develop and test
Options considered	Layered architecture, Pipe and filter architecture, client server architecture
Decision	Use a layered architecture to separate image capture, image processing, digit recognition, and results display
Consequences	Favors maintainability, accuracy, and readability by allowing each layer to be developed and tested independently. The cost is additional complexity in defining communication between layers, and changes to one layer may affect others.
Evidence	The architecture will be evaluated during implementation and testing
Related	QR- 01, QR-03, QR-04

ADR-02 Separate image preprocessing from digit recognition

Status	Accepted
Date and owner	November 8, 2026
Context (drivers)	QR-01, QR-03. Uploaded images may have different sizes, lighting conditions, or levels of clarity. The system needs to prepare images before sending them to the digit recognition component
Options considered	Perform preprocessing and recognition in one combined component. Use separate preprocessing and recognition components.
Decision	Use separate components for image preprocessing and digit recognition so that images are prepared before recognition begins
Consequences	Favors accuracy, modularity, and easier testing because preprocessing can be adjusted without rewriting the recognition logic. The cost is additional processing steps and the need to define how processed images are transferred between components.
Evidence	Recognition accuracy will be evaluated using test images with different image conditions.

Related	QR-01, QR-03, ADR-01

ADR-03 Display recognition results after processing is complete

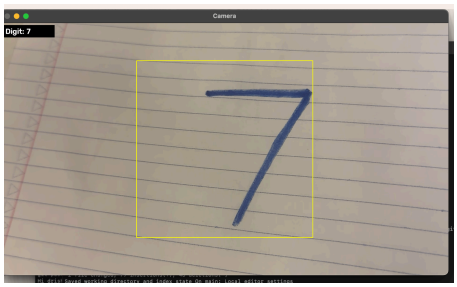
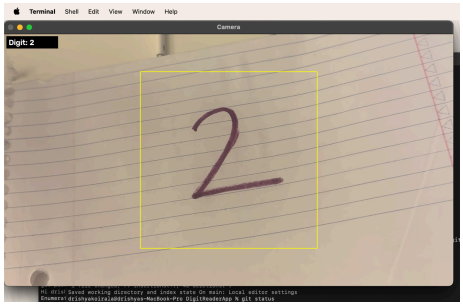
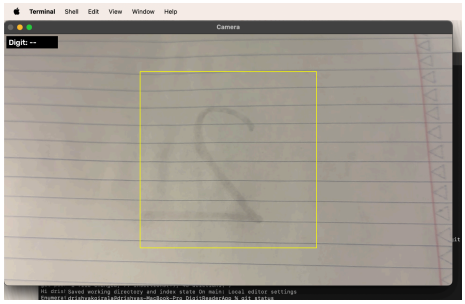
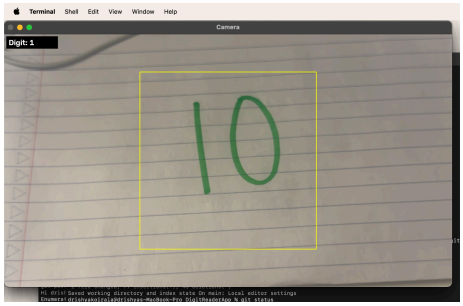
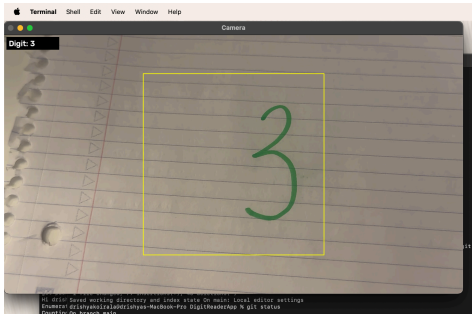
Status	Accepted
Date and owner	November 8, 2026
Context (drivers)	QR-04, QR-03. Users need to understand the digits recognized by the system and receive results in a clear and readable format after submitting an image.
Options considered	Display recognition results directly during processing. Display results in a separate results component after recognition is complete.
Decision	Use a separate results display component that presents the recognized digits after the image recognition process finishes.
Consequences	Favors usability and readability by keeping the user interface separate from the recognition logic. The cost is that users may need to wait until processing finishes before seeing results, and the system must pass recognition output to the display component.
Evidence	The results display will be evaluated through prototype testing
Related	QR-03, QR-04, ADR-01, ADR-02

7.2 Trade-off summary

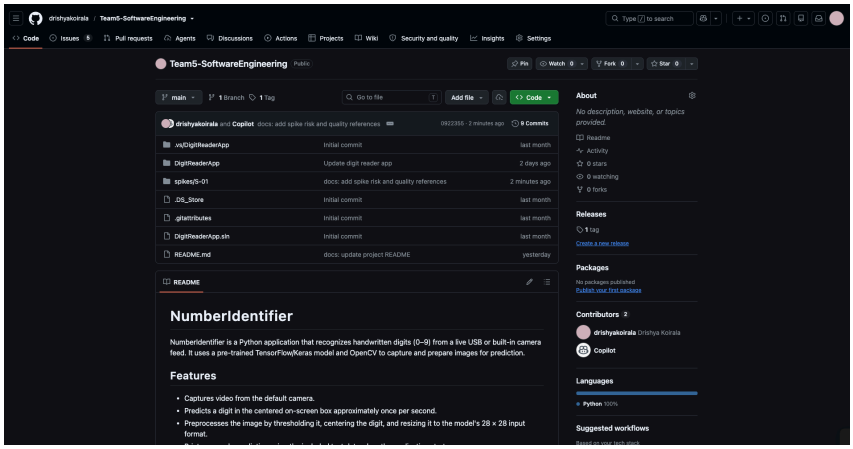
Quality favored	Architectural move	Cost accepted	Recorded in
Maintainability, accuracy, reliability	Use a layered architecture to separate system responsibilities	Additional complexity in communication between layers and possible dependencies when changes are made	ADR-01
Accuracy, modularity	Separate image preprocessing from digit recognition	Additional processing steps and communication between components	ADR-02
Usability, readability	Display recognition results through a separate results component after processing	Users must wait for processing to finish before receiving results	ADR-03

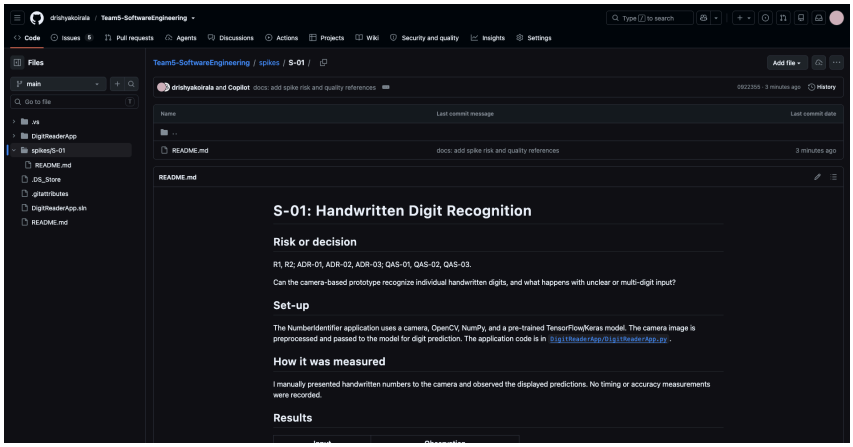
8 Prototype Evidence: Architecture Spike

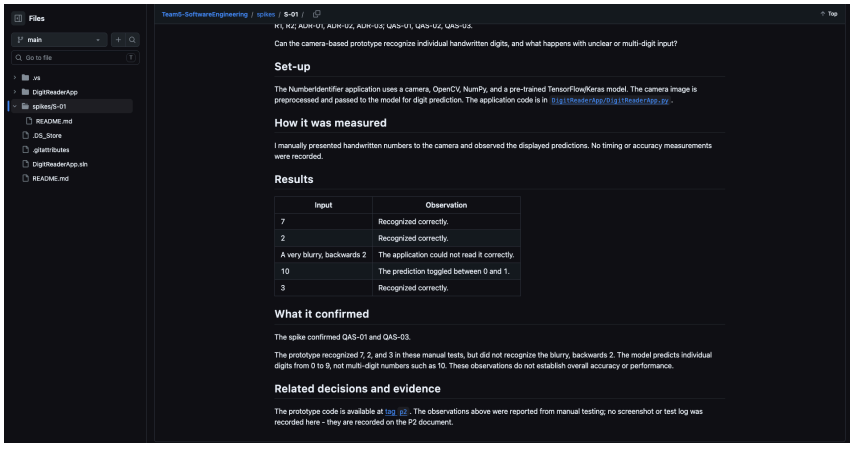
Spike ID and title	S-01 Handwritten Digit recognition test
Risk or decision it addresses	R1, R2 /ADR-01, ADR-02, ADR-03/ QAS-01, QAS-02, QAS-03
Question (answerable yes or no)	Can the program successfully recognize handwritten single-digit numerical responses? Yes
Set-up	Used the existing NumberIdentifier program and several examples of handwritten digits as input
How it was measured	Compared the program's predicted digits with the actual handwritten digits

Result (numbers)	    
What it changed or confirmed	QAS-01 and QAS-03 confirmed both of these QAS requirements.

Where to see it	Can see it on github under folder spikes/S-01







9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01(MUST) Scan single-digit input	UC-01	Teacher; imageInput	Image Capture	-	S-01 camera capture
FR-02(MUST) Recognize digits	UC-01; UC-03	ImageInput; processedImage;	Image Processing, Digit Recognition	-	S-01 recognized 7, 2 and 3
FR-03 (MUST) Record value	UC-01	RecognitionResult	Digit Recognition	-	Digit displayed. Recording unverified.
FR-04 Detect blanks	UC-01 UC-03	ProcessedImage, RecognitionResult	Image Processing	-	Empty field untested
FR-05 Flag uncertainty	UC-01 UC-02	RecognitionResult	Results Display	-	Review flag unverified
FR-06 Correct digit	UC-02	User RecognitionResult	Results Display	-	Manual correction unverified
QR-01(MUST) Accuracy	UC-01 UC-03	RecognitionModel RecognitionResult	Digit Recognition	QAS-01 ADR-01/02	S-01 examples only. No 100 response test
QR-02 Processing time	UC-01 UC-03	ImageInput ProcessedImage RecognitionResult	Image Processing Digit Recognition Results Display	-	No 50-field timing
QR-03(MUST) Display time	UC-01	RecognitionResult	Result Display	QAS-02 ADR-01/02/ 03	No 10 scan timing
QR-04(MUST) Readability	UC-01 UC-02	User RecognitionResult	Results Display	QAS-03 ADR-01/03	Labels visible Five-scan check pending
CON-01: Single digits only; no psychological interpretation	UC-01 UC-02	RecognitionModel; RecognitionResults	Digit Recognition	No explicit direct link	S-01 read 10 as 0 or 1

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
CON-02 Configured scale	UC-01/02/03	RecognitionResult(scale missing)	Owner pending	-	Scale validation untested

9.2 Gap analysis

Gap	Items	What you did about it
Requirements with no component	CON-02 lacks a scale-validation owner.	Marked ownership pending in 9.1
Components with no requirement, QAS or ADR	All four components link to requirements.	Mapped capture to FR-01, processing and recognition to FR-02, and display to QR-04
Use cases not analyzed (and why that is acceptable)	All three are expanded in 4.2, 5.2 covers UC-01 and UC-03	The two selected cases cover recognition and region setup. Logged the missing UC-03 diagram in 5.3

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

Team member	Role for P2 to P3	How the team makes a decision when it disagrees
Xiangrui Teng	Researcher	The team will first discuss disagreements openly and listen to each member's perspective. Then, we will make decisions based on the project requirements, available evidence, and what best meets our goals

10.2 Plan to P3

Milestone	Date	Owner	Done when
Complete and verify the existing code that reads a single handwritten digit	11/07	Prakriti Mehta	11/14
Increase reading accuracy to 95%	11/07	Telile Tura	11/21
Ability to read double digits	11/07	Drishya Koirala	11/28
Test the complete program, fix bugs, and finalize the application.	11/07	Sho Shimizu	12/04

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1 Poor image quality	Status remains the same, as poor image quality is user error.	Likelihood: 5 Impact: 9	Drishya Koirala - prompting user to retake image
R2 Indecipherable handwriting	The program will try its best to decipher, however, correct reading is not guaranteed. We still need to raise our confidence levels.	Likelihood: 5 Impact: 8	Telile Tura – include poor handwriting in model training
R3 May misread visually similar characters	This is dependent on the confidence score, and the ability to interpret we have increased accuracy of visually similar characters but we still need to perform testing.	Likelihood: 3 Impact: 9	Prakriti Mehta- train the model to decipher between different digit characters

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R4 Model limitations	Status remains the same, as model limitations are constrained by model capability. Thus, it is still a possibility based on what the user needs in a product.	Likelihood: 3 Impact: 4	Mehrsa Hosseini - be open with users regarding what our current model can do and how well it performs--work towards making improvements
R5 Slow runtime	Status remains the same: as error is still a possibility	Likelihood: 2 Impact: 3	Sho Shimizu - Remove unused training data

10.4 Repository

https://github.com/drishyakoirala/Team5-SoftwareEngineering

10.5 Use of AI tools in P2

Member	Tool	What it was used for	What you checked or changed
Drishya Koirala	Copilot	Programming	Each time I used copilot to code something I would check through the entire program to make sure that it makes sense to me such that it is implemented in the most efficient way. I also used copilot to commit and push my changes.

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q01	Will the program recognize multiple handwritten digits in one scan or only one digit at a time?	Drishya Koirala	Drishya Koirala 12/02	As for now, we have only tested with one digit in the given area that it captures.	The program will not be able to give a deterministic answer about the digit.
Q02	Will users take pictures directly through software or upload existing images?	Telile Tura	Telile Tura 12/01	Users will take pictures directly through software	Users have access to a webcam or a compatible device that the software has access to.

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q03	How should the system handle images that are blurry, distorted, or have poor lighting conditions that may affect digit recognition accuracy?	Prakriti Mehta	Prakriti Mehta 12/03	The program will attempt to recognize digits even if image is blurry, distorted, or poorly lit.	System may produce inaccurate predictions when image quality is poor
Q04	What should the program do when the camera captures an image but no handwritten digit is detected in the designated response area?	Xiangrui Teng	Xiangrui Teng 12/04	The program should indicate that no digit was detected	The program will display a message prompting the user to write a digit clearly and try again.
Q05	What should app do when there's no camera connected	Sho Shimizu	Sho Shimizu 12/04	Camera loss is handled as same as no digit	Show no camera found to window and retry every few second
Q06	Will the program be able to detect a number if the background contrast is low to the color of the number?	Mehrsa Hosseini	Mehrsa Hosseini 12/01	The threshold for detection is unknown	The program will need noticeable contrast between the number/s and their background

12 Baseline v1.0 Declaration (was P1 section 6)

Team member	Signature or initials	Date
Drishya Koirala	Drishya Koirala	11/7/2026
Prakriti Mehta	Prakriti Mehta	11/7/2026
Sho Shimizu	Sho Shimizu	11/7/2026
Telile Tura	Telile Tura	11/7/2026
Mehrsa Hosseini	Mehrsa Hosseini	11/8/2026
Xiangrui Teng	Xiangrui Teng	11/8/2026